

BISWAKETAN

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PROFESSIONAL SUMMARY

Engineer with hands-on experience building software from scratch, maintaining legacy systems, and deploying production platforms across AWS and Azure. Proficient in PostgreSQL, SQL query optimization, Kubernetes, Docker, and time-series data concepts. Comfortable across the full stack — from database internals to cloud deployment — with a proven ability to debug across layers, communicate technical findings clearly, and pick up new technologies fast. Takes ownership of problems and delivers without waiting to be directed.

TECHNICAL SKILLS

Databases: PostgreSQL · MySQL · SQLite · MongoDB · Neo4j · Redis · AWS Neptune

SQL & Query Optimization: Complex queries · Window functions · CTEs · PL/pgSQL · Execution plan analysis · Index tuning

Time-Series & Analytics: TimescaleDB · Hypertables · Continuous aggregates · Time-partitioned data · Real-time analytics

Cloud & Infrastructure: AWS (Lambda, Rekognition, Neptune, S3) · Azure · Kubernetes · Docker · GitHub Actions · GitLab CI

ML & AI Stack: Python · TensorFlow · PyTorch · Scikit-learn · Pandas · NumPy · LangGraph · CrewAI · Airflow · SHAP

Languages: Python · Go · TypeScript · JavaScript · Rust · C · C++ · Java · Bash

EXPERIENCE

AI/ML Engineer Intern

Jul 2025 – Sept 2025

CSM Technologies · Bhubaneswar, Odisha

- Built a demand forecasting service from scratch — designed the data pipeline, optimized SQL queries against relational databases, trained a TensorFlow model, and deployed predictions via a production REST API; maintained and refactored legacy scripts to improve reliability without breaking existing integrations.
- Containerized the inference service with Docker, deployed to cloud infrastructure, and debugged cross-layer failures across database queries, application logic, and API responses — reading logs and tracing root causes end-to-end.
- Translated model behavior and data quality findings into clear guidance for non-technical stakeholders, managing communication across the team throughout the project lifecycle.

PROJECTS

Multivariate Time Series Forecasting

Python · PostgreSQL · Scikit-learn · LSTM · Pandas · Matplotlib

- Built a forecasting pipeline over time-partitioned data in PostgreSQL, using CTEs and window functions to extract sequences, then trained and compared LSTM and gradient boosting models — analyzing seasonality, trend decomposition, and lag features across multiple correlated series.
- Visualized forecast confidence intervals and error metrics for business stakeholders, demonstrating ability to translate complex model outputs into actionable insight.

MLOps Pipeline using Apache Airflow

Python · Apache Airflow · Docker · PostgreSQL · CI/CD

- Orchestrated a full ML pipeline with Airflow — data ingestion, preprocessing, training, evaluation, and deployment — using PostgreSQL as the metadata store, with each stage containerized in Docker for environment consistency and reproducibility.
- Implemented DAG-based scheduling, failure alerting, and retry logic, building hands-on awareness of how production pipelines fail and how to trace failures across multiple services.

Multi-Agent System with LangGraph

Python · LangGraph · PostgreSQL · Agentic AI · REST API

- Built a multi-agent orchestration system where agents collaborate on complex tasks, with PostgreSQL as the backend for agent state, interaction history, and task records — designed for fast retrieval and full auditability.

- Debugged non-deterministic agent behavior by tracing execution logs and inspecting database state at each step, applying cross-layer debugging skills across application and data layers.

Fairness & Bias Audit Framework for ML Models

Python · Fairlearn · SHAP · Responsible AI

- Designed a bias auditing framework for loan approval models — measured demographic disparate impact with Fairlearn, applied SHAP for explainability, and implemented fairness-aware retraining across protected groups.
- Documented the audit process for team-wide reuse, demonstrating a habit of structured technical documentation and knowledge sharing.

EDUCATION

Bachelor of Computer Applications

2023 – 2026

KIIT University · Bhubaneswar, Odisha

Relevant coursework: Data Structures & Algorithms · DBMS · Cloud Application Development · Software Engineering

Class XII — PCM + Computer Science

2021 – 2023

Delhi Public School Kalinga

ACTIVITIES & LEADERSHIP

- Google GDC Campus Accomplishment Certificate
- Social Media Lead — KIIT University Page
- Student Organizer — OCIT | 23rd International Conference on Information Technology; contributed to planning and execution of an international academic conference hosted at KIIT University.